

Distilling Native Query–Key Responses into Compact Memory Landmarks

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Abstract

Progressive Retrieval Attention (PRA) uses compact routing indexes to choose long-memory chunks, then materializes the selected chunks’ unchanged native key/value (K/V) states. We ask whether low-rank native query–key interactions recover address signals erased by one mean gist. An initial gated study and a 120-controller sweep fail a prespecified response-distillation gate, but expose an evidence-supervised rank-16 QASPER frontier. We freeze that configuration and evaluate 64 fresh HotpotQA and 64 fresh QASPER identities with five seeds and a fixed four-chunk budget. On QASPER, rank-16 recall is 0.258 versus 0.117 for a native mean (paired +0.141, 95% interval [0.080, 0.210]); on HotpotQA it is 0.120 versus 0.125, while BM25 reaches 0.386. A further 100+100 identity study finds architecture transfer rather than universality: on 2WikiMultiHopQA, zero-shot and retrained rank 16 improve over mean by 0.033 and 0.093, while rank-8/eight-centroid routing retains 84.9% of the latter gain at 256 FP32 bytes/chunk. On MuSiQue, low-rank gains are unresolved; lexical routing remains stronger, and validation-selected lexical–low-rank fusion regresses on both new test cohorts. A Llama-family SmolLM2 replication preserves the QASPER-over-mean effect, while native-K/V generation shows that informed routing can still fail under a frozen late-layer consumer. Compact native interaction subspaces are a reusable QASPER/2Wiki routing channel, not a universal response compressor, lexical replacement, or complete memory-use solution.

1 Introduction

PRA separates *routing* from *reading*. A small summary chooses memory chunks; only the chosen chunks contribute their original token K/V to attention. The separation preserves pretrained values and bounds active memory, but it makes routing-summary quality consequential. One mean vector per chunk is attractive because it is deterministic, cacheable, and cheap. It is also a severe compression: a sparse key that responds strongly to a future query can disappear inside 31 unrelated tokens.

Earlier work already tested natural repairs. Multi-gist, prototype, k -means, SOM, local-gist, native-QK closure, lexical, and graph-query variants sometimes helped a dataset or controlled fixture, but did not establish a uniform natural frontier. Those methods mainly organize semantic or local geometry. This paper asks a different question: can a compact native-key set preserve the *response surface* that the frozen attention layer assigns to the full chunk?

The distinction matters. A key is useful because of how plausible queries address it, not because it reconstructs the mean of its neighbors or reads well as text. We therefore use full token-level keys as a teacher, measure ranking preservation directly, and keep the downstream materialization budget fixed. The study is gated. We first verify inherited Paper 2.6 rows, then test whether the teacher is meaningful, then ask whether a query-aware oracle has headroom. A learned selector is permitted only after those checks; more expressive off-manifold slots and recurrent memory remain closed if it fails.

The contributions are:

1. a tensor-native definition of chunk response distillation under grouped-query attention, with max, top- r , log-sum-exp, and attention-mass teachers;
2. matched native-key controllers, response metrics, routing metrics, changed-selection audits, and synchronized CUDA cost measurements;
3. a controlled 32–256-token suite and a frozen natural evaluation aligned with the Paper 2.5/2.6 32-token chunk and four-chunk budget;
4. a measured boundary: oracle landmarks show natural QASPER headroom, while a 321-parameter key-only selector recovers too little of that headroom to pass deployment gates;
5. a 120-controller query-conditioned sweep separating a prespecified primary result from test-selected exploratory findings and showing that response preservation, evidence ranking, and dataset utility are not interchangeable;
6. a post-G3 diagnostic frontier that crosses matched native geometry with direct rank-8/16/32 native-key projections, four/eight static low-rank representatives, and measured routing-index bytes and latency without changing native-K/V materialization;
7. an identity-disjoint 2WikiMultiHopQA/MuSiQue extension separating frozen zero-shot transfer, dataset-specific projection retraining, balanced four-dataset training, and validation-selected lexical-low-rank fusion under matched budgets;
8. a 128-identity confirmation with native-K/V generation and causal controls, a Llama-family/tokenizer replication, and a quantized vectorized serving index.

2 Native response distillation

2.1 Full-key teacher

Let chunk C contain pre-RoPE native keys $K_C \in \mathbb{R}^{n \times H_{kv} \times d_h}$ and let a frozen decoder produce query heads $q \in \mathbb{R}^{H_q \times d_h}$. The grouped-query map $g(h)$ assigns each query head to its native key head. Scaled token responses are

$$a_{i,h}(q, C) = \frac{q_h^\top k_{i,g(h)}}{\sqrt{d_h}}. \quad (1)$$

The teacher reduces tokens per head and then averages heads:

$$S^*(q, C) = \frac{1}{H_q} \sum_h F(a_{1,h}, \dots, a_{n,h}). \quad (2)$$

We validate four reductions: maximum, top- r mean, length-normalized log-sum-exp, and a softmax attention-mass proxy. Validation selects top-four mean with $r = 4$. This selection occurs before test evaluation and is frozen for every controller.

For compact landmarks $G(C) = \{\hat{k}_1, \dots, \hat{k}_m\}$, the estimate $\hat{S}(q, C)$ applies the same F and grouped-head mapping. The primary target is not $\|K_C - \hat{K}_C\|$. It is agreement between S^* and $\hat{S}$ over candidate chunks: score error, rank correlation, teacher top- k overlap, and KL divergence between softmax-normalized chunk distributions.

2.2 Controllers

All deployable landmark outputs are actual pretrained keys. Mean is the only synthetic control.

Mean Average valid keys independently in each native key head, yielding one key per chunk.

Last and random

Retain the final m valid keys or a deterministic random subset.

Farthest-first

Flatten native heads per token and greedily retain cosine-diverse keys.

Greedy oracle

Add the token that most reduces full-response squared error. It sees the evaluation query and is explicitly non-deployable.

Learned selector

Score each token from overall key norm, mean head norm, head-norm variation, cosine to the chunk centroid, local key change, forward/reverse position, and validity. A two-layer MLP, $8 \rightarrow 32 \rightarrow 1$, has 321 parameters. Hard top- m is used at inference.

Oracle selections for $m = 8$ supervise the learned model. Five seeds train for 400 AdamW steps on 16 validation identities. The backbone, native projections, tokenizer, and all natural test identities remain frozen. The selector receives no question vector, evidence label, dataset identifier, lexical feature, or chunk-level answer supervision at inference.

2.3 Query-conditioned continuation

The continuation addresses the key-only selector’s missing information need without changing native keys. Let $f_i \in \mathbb{R}^8$ be token i ’s normalized key-only features and let $\bar{q} \in \mathbb{R}^{H_q d_h}$ be the flattened, RMS-scaled native query. The selector logit is

$$z_i = \text{MLP}(f_i) + \frac{(W_q \bar{q})^\top (W_f f_i)}{\sqrt{r}}, \quad (3)$$

where $r \in \{8, 16, 32\}$. The resulting models have 16,769, 33,217, and 66,113 parameters. Hard top- m selects actual native keys at inference.

Training uses a piecewise-differentiable surrogate only to move selector logits. Within each chunk, the hard top- m support is retained while a softmax over its logits weights scalar native token responses. This gives a chunk surrogate $\tilde{S}(q, C)$. Final evaluation discards the surrogate, gathers hard-selected keys, and recomputes $\hat{S}(q, C)$ with the exact grouped-query native-QK function.

Four objectives are prespecified. *Oracle imitation* applies token-level weighted binary cross-entropy to greedy response-preserving selections. *Listwise* minimizes cross-entropy from the chunk softmax to a uniform distribution over evidence chunks. *Combined* adds $\text{KL}(\text{softmax } S^* \| \text{softmax } \tilde{S})$ to listwise ranking. *Decision-aware* adds pairwise softplus penalties between evidence chunks and non-evidence chunks nearest the fourth-place routing threshold. We cross all four objectives with both landmark counts, all three ranks, and five seeds: 120 controllers, each trained for 300 AdamW steps on validation only. Combined $r = 16, m = 4$ is the prespecified primary configuration; every other test comparison is reported without replacing it.

2.4 Post-G3 native geometry and low-rank routing

The original preregistration stops at G3. A separate diagnostic extension asks whether simple multimodal geometry explains the landmark result and whether routing needs full native-key width. First, deterministic Euclidean k -means centroids, cluster medoids, and farthest-first actual keys are constructed inside every 32-token chunk for $m \in \{1, 2, 4, 8\}$. Centroids are routing-only; selected chunks still materialize their unchanged token K/V. These controls use the same top-four-mean response reduction as the frozen teacher.

For direct low-rank routing, flatten the eight native key heads into $u_i \in \mathbb{R}^{1024}$ and the 16 native query heads into $\bar{q} \in \mathbb{R}^{2048}$. A validation-derived scalar RMS normalizes keys without changing relative token norms. The learned score is

$$\ell_i = \frac{(W_q \bar{q})^\top (W_k u_i)}{\sqrt{r}}, \quad W_q \in \mathbb{R}^{r \times 2048}, \quad W_k \in \mathbb{R}^{r \times 1024}, \quad (4)$$

for $r \in \{8, 16, 32\}$. The models contain 24,576, 49,152, and 98,304 parameters. Combined listwise evidence ranking and native-response distillation train the projections for 300 steps on validation identities. The backbone, native projections, and test identities remain frozen.

At index-construction time, $z_i = W_k u_i$ is cached. Online work is $z_q = W_q \bar{q}$ followed by rank- r dots; no deployable low-rank row scans full native keys. We evaluate all 32 projected tokens and static k -means, medoid, and farthest-first indexes with $m \in \{4, 8\}$. Thus token compression ($32 \rightarrow m$) and interaction compression ($1024 \rightarrow r$) are measured separately and jointly. With FP32 routing indexes, a chunk uses 131,072 bytes for 32×1024 full native keys, 4,096 for one mean key, $4mr$ for static low-rank representatives, and $128r$ for all projected tokens.

This accounting separates routing-index residency from backing native K/V, selected-K/V transfer, and active materialization. Query-conditioned native landmark selection also needs its full token selector index because landmark identities vary with the query; it cannot honestly be counted as only m persistent native keys. Construction time, cached online latency, native dots, low-rank dots, and eager peak-GPU deltas are recorded independently.

2.5 Matched materialization

The intervention changes only routing summaries. Candidate units are the same 32-token local chunks. Every method requests four chunks, so it materializes approximately 128 native K/V tokens: 126.1 on HotpotQA because of short final chunks and 128.0 on QASPER. Active memory fractions are 8.8% and 3.3%, respectively. Values are never compressed, projected, or synthesized.

3 Protocol

3.1 Frozen natural cohort

We reuse Qwen3-0.6B revision c1899de2, layer 27, 16 query heads, eight K/V heads, and head width 128. Source blocks are encoded in bounded 256-token calls, then divided into 32-token routing chunks without local re-encoding. Pre-RoPE keys and attention-input query states preserve token spans and evidence masks. The query state is projected through the unchanged native `q.proj` and `q.norm`.

Validation contains eight HotpotQA and eight QASPER identities. Test contains 16 of each, matching the frozen Paper 2.5/2.6 identities and excluding validation identities. Feature tensors are 329 MB and 621 MB, respectively; manifests record SHA-256 hashes. Paper 2.6’s 224 test rows are re-aggregated exactly at G0. Paper 2.5’s 256-token-parent, 20%-budget rows are retained only as historical context because their candidate unit and budget are not numerically matched.

3.2 Cross-dataset transfer cohort

The extension reuses Paper 2.7’s separately frozen natural identities: each of 2WikiMultiHopQA and MuSiQue contributes 20 validation and 100 test questions [7, 8]. The 200 test identities are disjoint from validation and from the historical 74-query cohort. 2WikiMultiHopQA supporting-fact sentences and MuSiQue supporting paragraphs supply evidence; answers and model outputs never define labels. Unique document/sentence markers make each authored evidence item an exact source substring before fast-tokenizer alignment. Manifests record source hashes, identity hashes, candidate counts, evidence counts, model/tokenizer revision, and feature-cache hashes.

The model contract remains Qwen3-0.6B revision c1899de2, layer 27, pre-RoPE native keys, 16 query heads, eight K/V heads, head width 128, 256-token bounded source calls, and 32-token routing chunks. Sources contain 315–4,897 tokens but no model call exceeds 256 source tokens. The backbone and backing native K/V stay frozen.

Three transfer regimes share the fixed rank-8/rank-16 architecture and combined ranking–response objective. *Zero-shot* applies the inherited Hotpot/QASPER projection unchanged. *Retrained* fits only W_q, W_k on the new dataset’s 20 validation identities. *Mixed* balances 20 samples from each of Hotpot, QASPER, 2Wiki, and MuSiQue by cycling the smaller inherited validation cohorts. Each regime uses five seeds; neither rank nor objective is selected from new test identities.

All examples have two to four authored evidence items, so four selected chunks can in principle touch every chain item. Full supporting spans are longer: they occupy 3–60 physical chunks, and only 11.7% fit completely in four chunks. We therefore retain $K = 4$ as primary and add validation-fixed $K = 6$ for 2WikiMultiHopQA and bounded $K = 8$ for MuSiQue. The latter rows are sensitivity analyses, not claims that every supporting token can be materialized. Evidence recall counts positive physical chunks; evidence-group completion asks whether at least one chunk from every contiguous authored evidence region is selected.

3.3 Controlled suite

Twenty synthetic cases cross chunk sizes $n \in \{32, 64, 128, 256\}$. Each case has 16 candidate chunks. An evidence chunk contains sparse query-aligned native keys while a distractor has a coherent but weaker mean. This tests whether full-key and landmark controllers preserve a sparse address that averaging can hide. The suite is a mechanism check, not evidence for natural language performance.

3.4 Metrics and uncertainty

Native preservation metrics are MAE, RMSE, Spearman ρ , Kendall τ , KL divergence, and teacher top- k overlap for $k \in \{1, 2, 4, 8\}$. Routing metrics are evidence recall and precision, any-evidence recall, contiguous evidence-group completion, exact selected identity, MRR, active memory fraction, evidence-normalized overhead, materialized tokens, native dots, and wall time. Five-thousand-draw paired bootstrap intervals resample identities. Selector seeds are averaged within identity before comparisons with deterministic baselines.

The prespecified gates are:

- G0: inherited Paper 2.6 aggregation parity;
- G1: full-K beats mean on controlled recall and at least one natural dataset;
- G2: greedy $m \leq 8$ improves teacher top-four overlap and at least one natural recall result without changing materialized K/V;
- G3: the learned selector recovers at least 80% of oracle gain, improves recall or completion on one natural dataset, and regresses by less than 0.02 elsewhere;

Table 1: Held-out four-chunk routing. Learned rows average five seeds. Oracle rows use the test query and are non-deployable. Values are evidence recall; parentheses give teacher top-four overlap.

Method	HotpotQA	QASPER
Mean key, $m = 1$	0.124 (0.594)	0.098 (0.516)
Full-K teacher	0.124 (1.000)	0.125 (1.000)
Farthest-first, $m = 2$	0.142 (0.734)	0.130 (0.422)
Greedy oracle, $m = 4$	0.145 (0.922)	0.178 (0.703)
Greedy oracle, $m = 8$	0.124 (0.984)	0.136 (0.812)
Learned selector, $m = 4$	0.122 (0.741)	0.083 (0.422)
Learned selector, $m = 8$	0.124 (0.834)	0.139 (0.553)

- G4/G5: synthetic slots and streaming memory run only after G3 passes.

The post-G3 extension uses E0–E6 and does not revise G3. E0 verifies frozen artifact, cache, model, and identity parity. E1 compares query/response-aware selection with matched native geometry. E2 applies a cross-dataset non-regression rule to query conditioning; test-selected rows remain exploratory. E3 compares ranking/decision objectives with oracle imitation. E4 requires at least 95% retention of the best full-dimensional retrieval gain on a natural dataset and at least $4\times$ less routing state. E5 asks whether joint m, r compression matches the QASPER semantic gist on a better quality-bytes-latency point. E6 opens 2Wiki/MuSiQue only if E2–E5 form a credible frontier.

The reopened cross-dataset study uses M0–M5 without revising G3 or E0–E6. M0 audits frozen rows, feature semantics, identity separation, and evidence alignment. M1 requires rank-16 to improve over mean with a paired interval excluding zero, or an explicitly material effect with strong directional consistency. M2 asks whether rank-8/eight-centroid routing retains at least 70% of the rank-16 gain while using at least $8\times$ less routing state than mean. M3 requires lexical-low-rank fusion to beat the best validation-selected lexical channel at fixed K . M4 requires a useful unchanged architecture family on at least two of QASPER, 2Wiki, and MuSiQue; dataset-specific weights are permitted. M5 opens native-K/V generation only for a dataset that passes M1 or M3.

4 Results

4.1 Teacher sanity and oracle headroom

G0 passes with zero mismatches across 224 inherited rows. Validation selects top-four mean across query heads: its validation evidence recall is 0.1387 versus 0.1255 for the corresponding mean-key control. Max, log-sum-exp, and attention-mass teachers do not beat that control on validation.

G1 also passes, narrowly on natural data. Full-K recall is 1.000 versus 0.950 for mean in the controlled suite. On held-out HotpotQA it is 0.1240 versus 0.1235; on QASPER it is 0.1252 versus 0.0976. The natural teacher is therefore not a strong retrieval oracle by itself. It provides a modest QASPER signal and essentially no Hotpot gain.

G2 passes. Relative to mean, greedy $m = 8$ improves aggregate teacher top-four overlap by 0.3438. Recall is not monotone in m : $m = 4$ is best, improving HotpotQA by 0.0219 and QASPER by 0.0802. The paired intervals, $[-0.0313, 0.0750]$ and $[-0.0142, 0.2233]$, both include zero. Oracle landmarks therefore establish response-preservation headroom and a promising QASPER direction, not a statistically resolved natural retrieval gain.

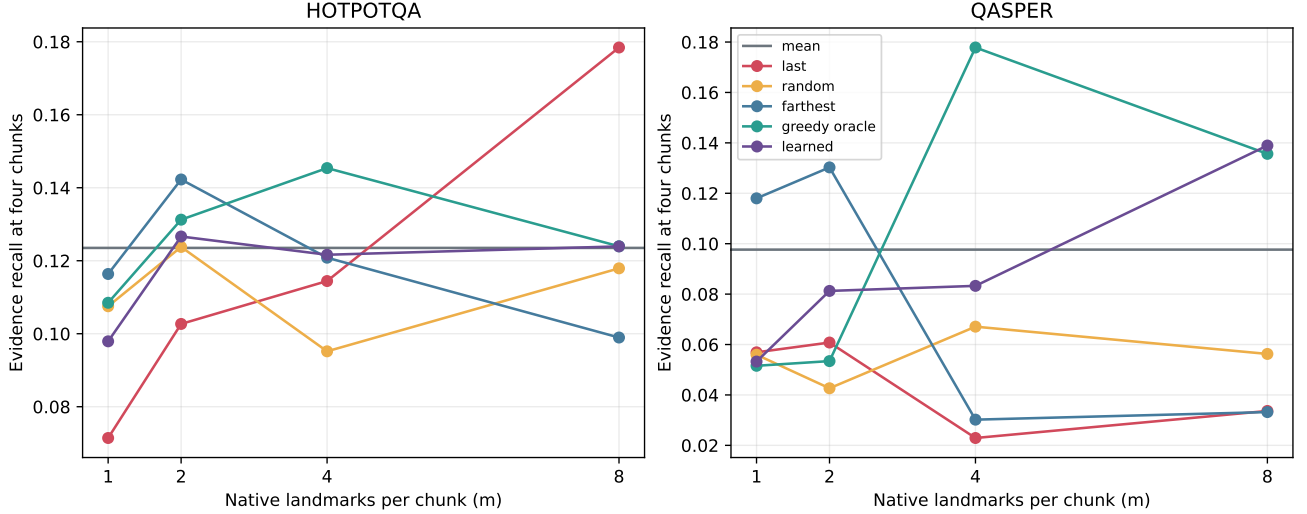


Figure 1: Natural evidence recall under the matched four-chunk materialization budget. The dataset-separated view exposes the Hotpot/QASPER asymmetry.

4.2 Learned selector: useful direction, failed gate

The learned selector does not recover the oracle. At $m = 8$, it captures 40.5% of the oracle teacher-overlap gain, below the 80% threshold. Its HotpotQA recall delta over the QK mean is +0.0004 with interval $[-0.0573, 0.0533]$. Its QASPER delta is +0.0413 with interval $[-0.0589, 0.1670]$. G3 therefore fails and G4/G5 do not run.

Seed variation is material. HotpotQA $m = 8$ recall ranges from 0.1173 to 0.1298. QASPER ranges from 0.0478 to 0.1821; four seeds are at or above 0.119, while seed 37 collapses. Teacher top-four overlap is more stable than evidence recall, showing that response imitation and evidence utility are related but not interchangeable.

4.3 Query conditioning: objective-dependent utility

The prespecified combined $r = 16, m = 4$ controller improves HotpotQA recall from the full-key teacher’s 0.1235 to 0.1667, a paired difference of 0.0432 with interval $[-0.0112, 0.0970]$. It also improves over the key-only $m = 8$ selector by 0.0427, interval $[-0.0015, 0.0907]$. This is the best HotpotQA result in the 24-configuration continuation, but remains below BM25, exact, and hybrid routing. The same controller reaches only 0.0574 on QASPER, 0.0402 below the mean-key control and 0.1202 below exact routing. Query conditioning therefore does not produce a dataset-uniform primary result.

The test-selected exploratory result is more encouraging but is not confirmatory. Decision-aware $r = 32, m = 8$ reaches 0.1718 QASPER recall, compared with 0.1776 for Paper 2.6 exact routing, 0.1389 for the key-only selector, and 0.0976 for the mean key. Its paired differences are -0.0059 against exact, interval $[-0.1831, 0.2049]$, and $+0.0328$ against key-only, interval $[-0.0292, 0.1073]$. Five seed means span 0.1639–0.1796. The narrow seed range shows optimization stability for this configuration; the wide identity-bootstrap intervals show that 16 test questions per dataset are still too few to resolve comparative retrieval utility.

Table 2: Matched four-chunk evidence recall. “Primary” was fixed before test evaluation. “Exploratory” is the best QASPER row selected after inspecting the 24-configuration test matrix.

Method	HotpotQA	QASPER
QK mean	0.124	0.098
Key-only learned, $m = 8$	0.124	0.139
Paper 2.6 BM25	0.398	0.042
Paper 2.6 exact	0.287	0.178
Paper 2.6 hybrid	0.279	0.081
Primary combined, $r = 16, m = 4$	0.167	0.057
Exploratory decision-aware, $r = 32, m = 8$	0.143	0.172

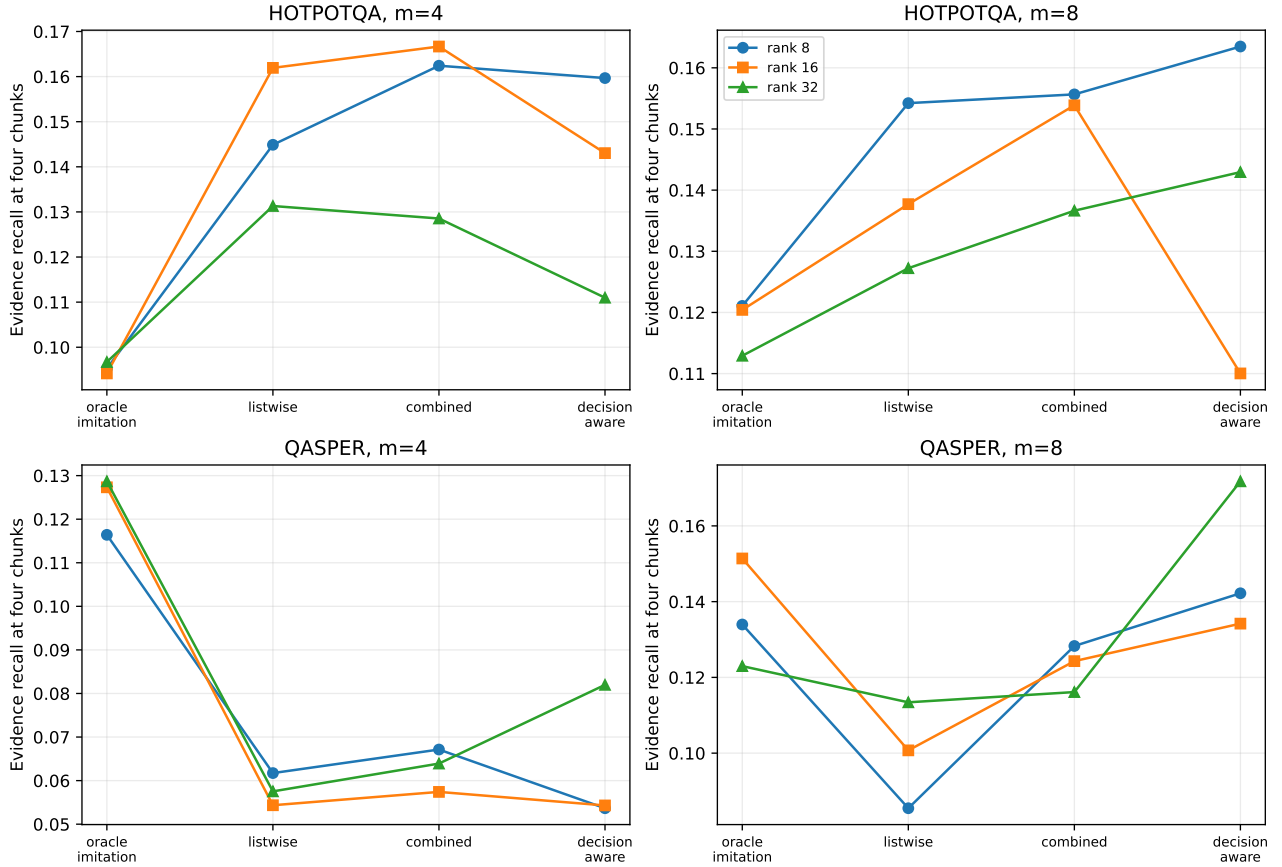


Figure 2: Evidence recall across objective, low-rank width, and landmark count. Objective choice changes both the level and the dataset direction; additional rank or landmarks are not monotonically beneficial. Points average five selector seeds.

Response preservation does not select the same controller as retrieval utility. Oracle-imitation $r = 32, m = 8$ gives the best aggregate teacher top-four overlap, 0.717 versus 0.555 for mean and 0.898 for greedy oracle, recovering 47.3% of oracle gain. Yet across the 24 configurations, teacher overlap and evidence recall correlate negatively on HotpotQA ($r = -0.665$) and positively on QASPER ($r = 0.682$). These descriptive correlations do not establish causality, but they make the failure mode visible: matching a frozen layer’s response can preserve preferences that are not evidence-bearing.

Table 3: Post-G3 diagnostic frontier. Bytes are persistent FP32 routing-index bytes per 32-token chunk; cached latency excludes index construction and native-K/V materialization. Low-rank rows average five seeds.

Method	Bytes/chunk	Cached ms	HotpotQA	QASPER
Native mean, $d = 1024$	4,096	1.11/1.19	0.124	0.098
Native farthest, $m = 2$	8,192	2.76/5.27	0.142	0.130
Low-rank all, $r = 8$	1,024	0.86/0.88	0.104	0.246
Low-rank all, $r = 16$	2,048	0.90/0.96	0.121	0.254
Low-rank all, $r = 32$	4,096	0.90/0.92	0.145	0.218
Low-rank k -means, $r = 8, m = 8$	256	0.90/0.96	0.100	0.183
Paper 2.6 exact	—	—	0.287	0.178

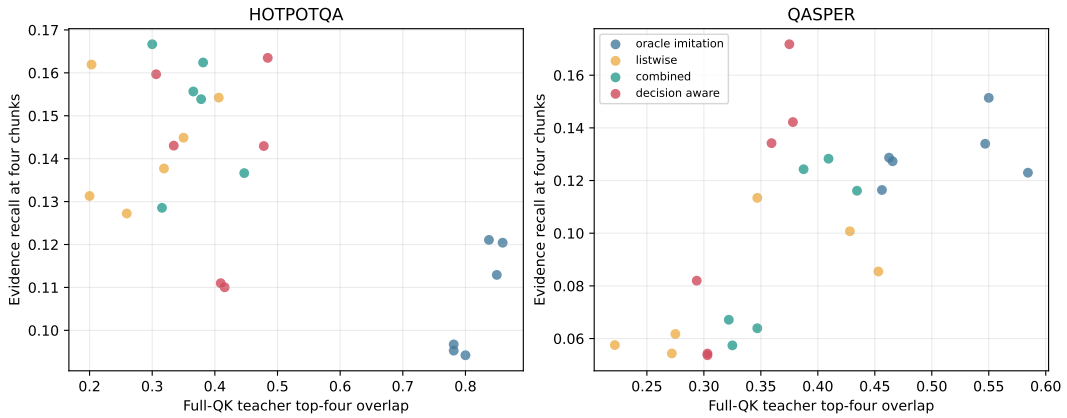


Figure 3: Response preservation versus evidence recall for all 24 query-conditioned configurations. The opposing dataset trends explain why the highest-fidelity compressor is not a universal retrieval controller.

The primary selector has 33,217 parameters and trains in 36.2 seconds per seed; the exploratory selector has 66,113 parameters and trains in 50.2 seconds per seed on the GTX 950M. Synchronized selector-plus-top- m latency is 42.6/117.0 ms for the primary and 42.9/120.5 ms for the exploratory controller on HotpotQA/QASPER. These continuation timings exclude compact-key QK scoring and materialization, so they are component measurements rather than direct replacements for Table 8.

4.4 Post-G3 low-rank frontier

Simple native geometry does not explain the continuation result. At matched $m = 2$, farthest-first is the strongest geometric control on both datasets, reaching 0.1423 HotpotQA and 0.1303 QASPER recall. Increasing to four or eight centroids, medoids, or farthest points is not monotone. In particular, four native centroids reach only 0.0332 on QASPER despite 0.453 teacher top-four overlap. Preserving several modes of native-key geometry is therefore insufficient by itself.

Direct low-rank native interaction gives a different result. Scoring all 32 projected tokens at rank 16 reaches 0.2542 QASPER recall, a gain of 0.1566 over the native mean with paired interval $[0.0462, 0.3001]$. It also exceeds the test-selected query-conditioned controller by 0.0825, interval $[0.0366, 0.1337]$. The gain over Paper 2.6 exact routing is 0.0766, but its interval $[-0.0791, 0.2763]$ includes zero. Five seed means are 0.1940–0.3027. Rank 8 is close at 0.2456; rank 32 falls to 0.2179 and has the largest seed standard deviation, 0.0666. Extra rank is not reliably useful.

The all-token rank-16 index is $64\times$ smaller than the 32×1024 full native-key stream and $2\times$ smaller than one full-width mean key. Joint token/rank compression is more aggressive: rank-8 k -means with eight representatives uses 256 bytes per chunk, $512\times$ below the full stream and $16\times$ below the mean, while reaching 0.1829 QASPER recall. Its gain over exact is only 0.0052 with interval $[-0.1250, 0.1675]$. Index construction is expensive in this reference implementation (2.26 seconds per QASPER example for rank-8 k -means), but it is offline and cacheable; cached online scoring is 0.96 ms. These timings are component measurements on the GTX 950M, not production throughput claims.

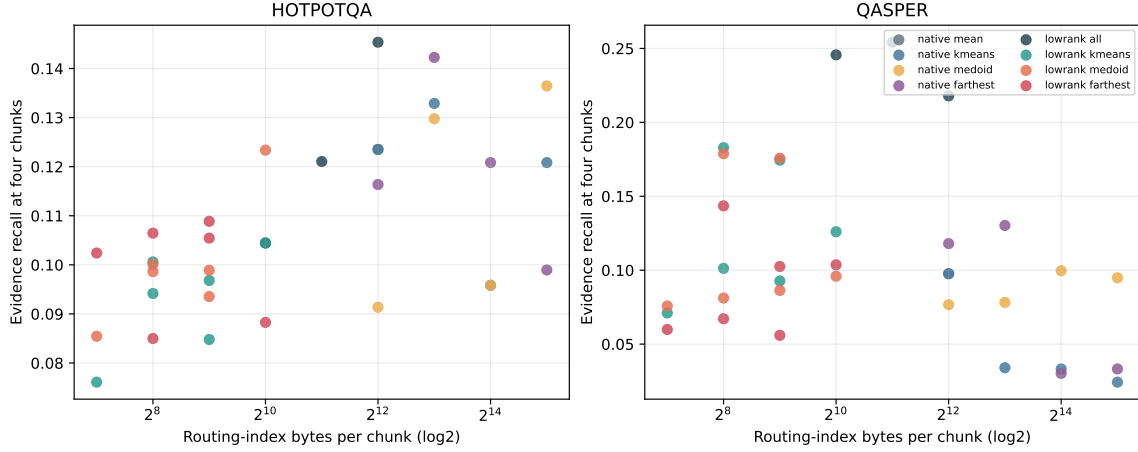


Figure 4: Evidence recall against persistent routing-index bytes per chunk. Direct low-rank routing creates a new QASPER quality–memory frontier; Hotpot remains dominated by lexical routing.

Low teacher overlap limits the interpretation. Rank-16 all-token routing overlaps the teacher’s top four by only 0.228 on QASPER and 0.309 on HotpotQA. The selected chunks differ from the mean-key selection for every test identity. This controller has learned an evidence-bearing interaction subspace from validation labels; it has not distilled the frozen teacher faithfully.

Structural ablations reinforce the dataset asymmetry. On HotpotQA, query-independent salience at $m = 4$ reaches 0.1813, above every bilinear variant in the ablation. On QASPER, bilinear-only rank 32, $m = 8$ reaches 0.1488, above its salience-plus-bilinear counterpart at 0.1161. These are test-matrix diagnostics, not prespecified winners. They show that query interaction contributes on QASPER while learned token importance can suffice on Hotpot; simply adding both terms does not guarantee a better selector.

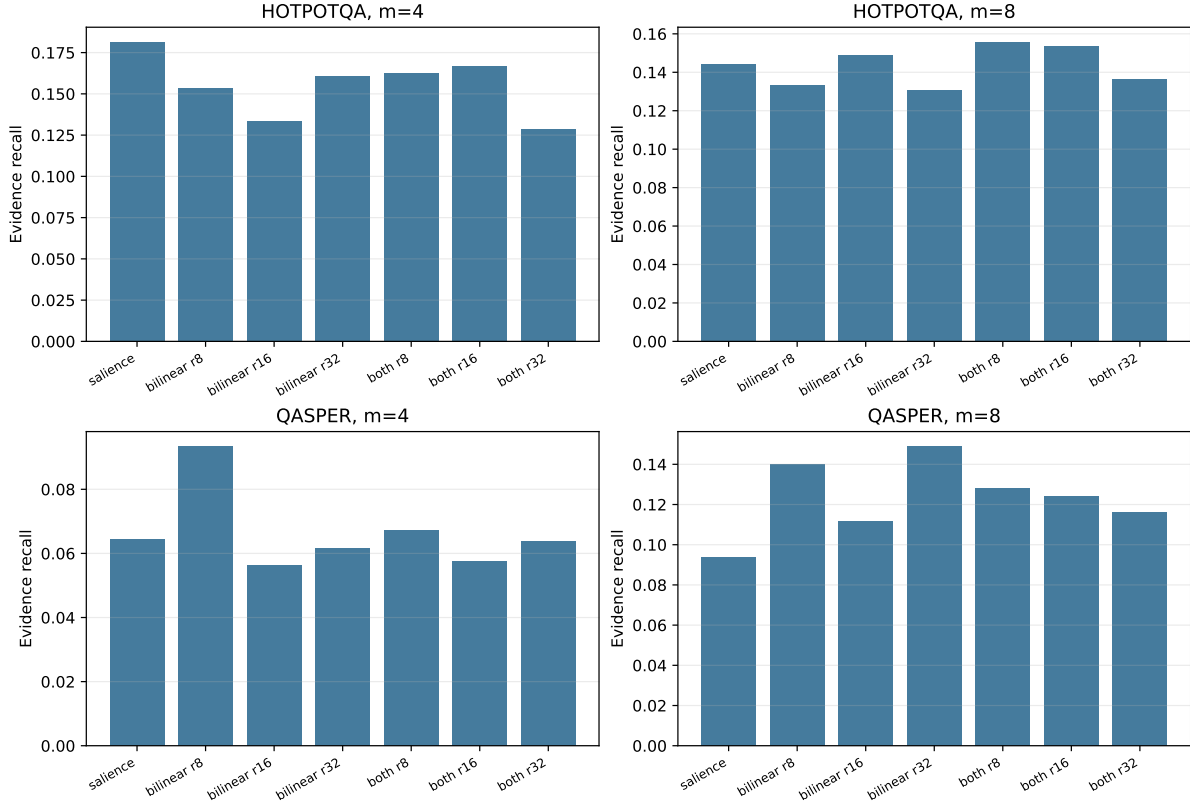


Figure 5: Selector structure ablation at four and eight landmarks. Saliency and bilinear query interaction have different dataset roles, and their combination is not uniformly additive. Points average five seeds.

4.5 Cross-dataset extension: 2WikiMultiHopQA and MuSiQue

The extension passes its feature audit: all 240 identities have exact authored-evidence substrings, validation and test are disjoint, and every source call respects the 256-token model bound. The primary $K = 4$ result is mixed. On 2WikiMultiHopQA, inherited rank 16 improves over mean by 0.0328, interval [0.0025, 0.0638]; dataset-specific retraining improves by 0.0926, interval [0.0582, 0.1289]; and balanced four-dataset training improves by 0.0777, interval [0.0407, 0.1128]. The unchanged architecture therefore transfers both weakly without new labels and more strongly after projection-only adaptation.

It does not become the best channel. Validation selects exact routing on 2WikiMultiHopQA, whose held-out recall is 0.282 versus 0.190 for retrained rank 16. The paired difference is -0.0918 , interval $[-0.1446, -0.0397]$. On MuSiQue, zero-shot, retrained, and mixed rank-16 effects over mean are -0.0010 , $+0.0043$, and $+0.0051$; every interval crosses zero. Validation selects BM25 at 0.122 recall, while held-out exact reaches 0.130 as a reported but test-unselected diagnostic. 2WikiMultiHopQA thus adds a second useful low-rank regime beside QASPER; MuSiQue behaves more like the lexical/entity-dominated Hotpot regime.

Table 4: Fresh cross-dataset routing at four chunks. R is physical evidence-chunk recall; C is contiguous authored evidence-group completion. Learned rows ensemble five seeds. Bold marks the highest reported held-out value, not a validation-selected replacement.

Method	2WikiMultiHopQA		MuSiQue	
	R	C	R	C
Native mean	.098	.10	.067	.13
Exact	.282	.18	.130	.24
BM25	.267	.24	.122	.20
Rank 16, zero-shot	.130	.10	.066	.11
Rank 16, retrained	.190	.18	.071	.15
Rank 16, mixed	.175	.20	.072	.15
Rank 8, eight centroids, retrained	.176	.15	.068	.14
Lexical + rank 16, retrained	.228	.21	.094	.14

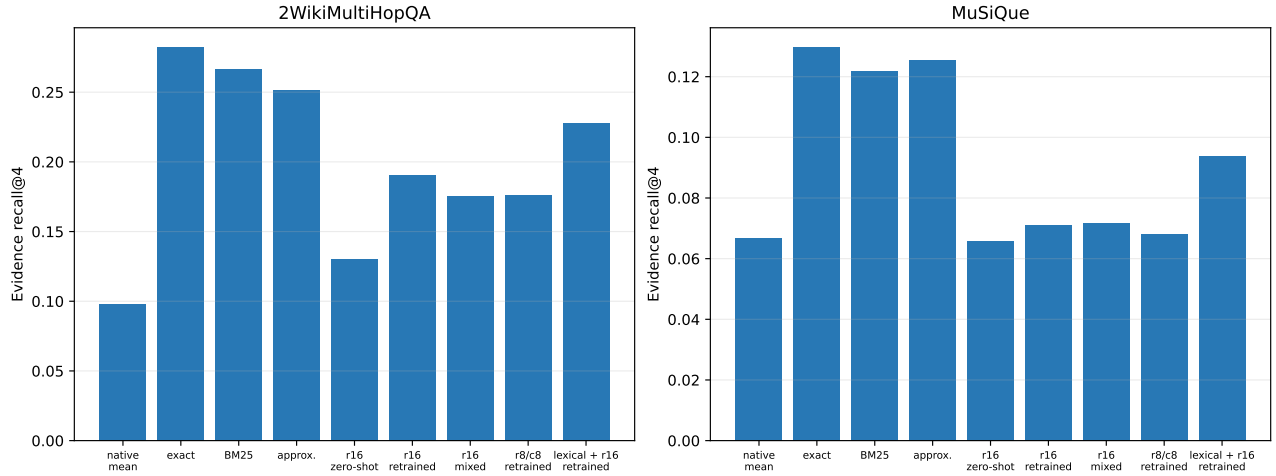


Figure 6: Fresh-cohort evidence recall at the fixed four-chunk budget. Low-rank transfer is resolved on 2Wiki, but exact routing remains stronger; MuSiQue remains lexical. The lexical-plus-rank-16 row uses validation-selected weights and illustrates failed calibration rather than an oracle fusion.

Compression survives where the native channel helps. On 2WikiMultiHopQA, rank-8/eight-centroid recall is 0.176; its gain over mean is 0.0786, interval $[0.0437, 0.1143]$. It retains 84.9% of the rank-16 gain while using 256 rather than 2,048 FP32 routing bytes/chunk and $16\times$ less state than a full-width mean. On MuSiQue the corresponding retention ratio is only 31.8%, because neither numerator nor rank-16 gain is resolved. M2 therefore passes on 2WikiMultiHopQA, not as a dataset-independent compression law.

More budget improves coverage without changing channel order. At $K = 6$, 2WikiMultiHopQA exact recall is 0.330 and retrained rank 16 is 0.274; their completion values are 0.26 and 0.34. At $K = 8$, MuSiQue’s validation-selected BM25 reaches 0.183 recall and 0.36 completion, versus 0.130 and 0.26 for retrained rank 16. These rows materialize about 191 and 255 tokens, respectively, and remain separate sensitivity analyses.

Static fusion fails M3. The retrained lexical–low-rank policy is selected at 0.928 validation recall on 2WikiMultiHopQA but falls to 0.228 on test, 0.0544 below exact with interval $[-0.0995, -0.0093]$. On MuSiQue it is selected at 0.627 validation recall and reaches 0.094 on test, 0.0281 below validation-selected BM25 with interval $[-0.0473, -0.0099]$. Selection disagreement is real, but not automatically

useful: retrained low rank adds 0.73/0.77 unique evidence chunks per 2Wiki/MuSiQue example relative to the selected lexical channel, while lexical routing adds 1.26/1.37. Under a fixed budget, complementary distractors and 20-example validation overfit outweigh the added evidence.

Seed behavior makes the same caution visible. Retrained rank-16 per-seed recall spans 0.115–0.186 on 2WikiMultiHopQA and 0.053–0.075 on MuSiQue; mean pairwise selected-set Jaccard is only 0.188 and 0.157. Score ensembling improves the headline 2Wiki row, but a deployed single index should not inherit ensemble cost silently. The serving audit below therefore fixes seed 11.

4.6 Fresh confirmation and causal memory use

We froze rank 16, rank-8/eight-centroid construction, five checkpoints, chunk size, and four-chunk budget before opening a new offset-24 cohort of 64 identities per dataset. No confirmation identity overlaps the earlier natural matrix. Table 5 shows that the low-rank effect is reproducible but dataset-specific. Rank 16 improves QASPER recall over mean by 0.1407, interval [0.0795, 0.2099], and over hybrid by 0.1149, interval [0.0562, 0.1796]. Its -0.0600 difference from exact remains unresolved. Neither low-rank index improves HotpotQA over mean, and all remain far below lexical channels.

Table 5: Fresh 64+64 confirmation, evidence recall at four chunks. Learned rows ensemble five validation-trained seeds.

Method	HotpotQA	QASPER
Native mean	0.125	0.117
BM25	0.386	0.065
Exact	0.282	0.318
Hybrid	0.224	0.143
Rank 16, all projected tokens	0.120	0.258
Rank 8, eight centroids	0.129	0.206
Evidence oracle	0.814	0.731

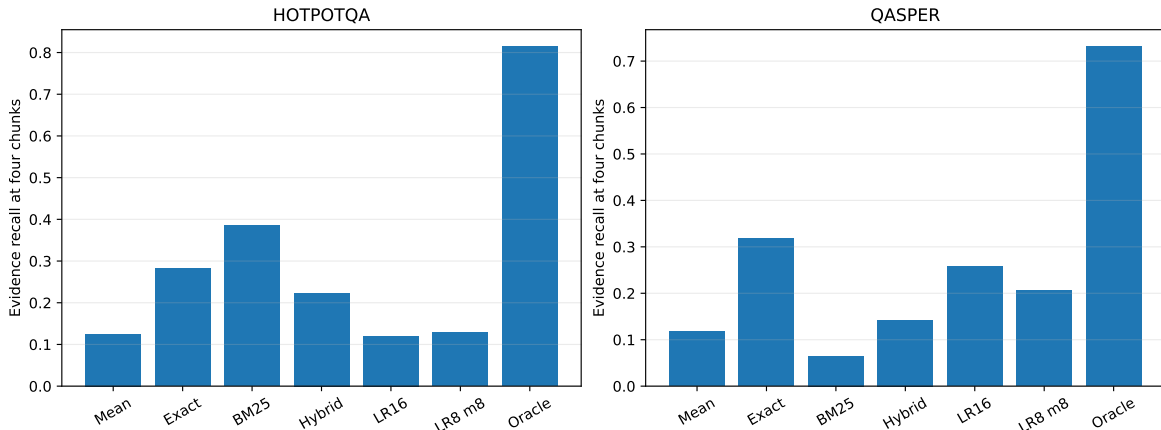


Figure 7: Fresh-cohort routing confirmation. The rank-16 gain transfers within QASPER, while Hotpot remains lexical.

We next replay the saved chunk identities through unchanged layer-27 native K/V. Teacher-forced gold-token log-probability is primary; 12-token greedy EM/F1 is secondary because the 0.6B frozen model rarely emits exact answers. Table 6 separates absolute memory use from routing quality. On HotpotQA, rank 16 improves over no memory by 0.0738, interval [0.0074, 0.1400], but differs from

shuffled memory by only 0.0091, with an interval crossing zero. On QASPER, rank 16 is -0.2124 versus no memory, interval $[-0.4079, -0.0194]$, yet exceeds shuffled memory by 0.3115, interval $[0.1378, 0.4887]$, and bottom-ranked memory by 0.4699, interval $[0.2369, 0.7118]$. Selection is causally better than wrong memory, but the frozen late-layer consumer does not turn the retrieval gain into a net benefit.

Table 6: Mean gold-token log-probability change relative to no memory. All memory rows request four chunks.

Condition	HotpotQA	QASPER
Native mean	+0.087	-0.095
Rank 16	+0.074	-0.212
Rank 8, eight centroids	+0.034	-0.194
Shuffled chunks	+0.065	-0.524
Bottom-ranked chunks	-0.035	-0.682
Evidence oracle	+0.076	-0.092
Direct evidence text	+2.956	+0.225

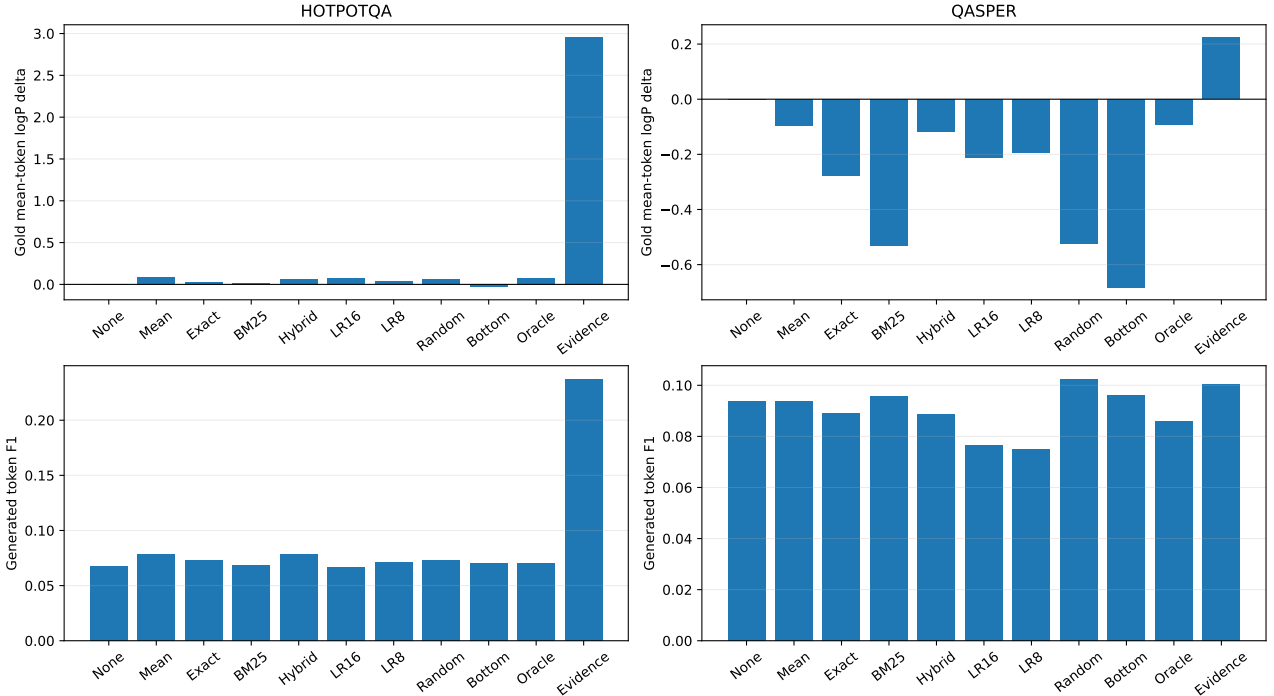


Figure 8: Native-K/V generation and causal controls. Routing quality and memory integration are distinct bottlenecks.

4.7 Cross-model replication

We repeat validation training and fresh-cohort routing on SmolLM2-135M, a Llama-family model with a different tokenizer, 30 layers, 9 query heads, 3 native K/V heads, and 64-wide heads. The capture uses layer 29 and the model’s own normalized pre-RoPE query/key projections. Rank-16 QASPER recall is 0.155 versus 0.092 for mean, a paired gain of 0.0638 with interval $[0.0192, 0.1148]$. Rank-8/eight-centroid recall is 0.165, gain 0.0739 with interval $[0.0226, 0.1342]$. Exact reaches 0.298 and remains significantly stronger. On HotpotQA, neither learned index improves over mean and BM25 again leads. The low-rank

QASPER signal is therefore portable across model family and tokenizer, but its channel ordering is also portable: exact lexical evidence remains the stronger control.

Table 7: SmolLM2/Llama-family fresh-cohort recall.

Method	HotpotQA	QASPER
Native mean	0.159	0.092
BM25	0.394	0.059
Exact	0.277	0.298
Hybrid	0.246	0.117
Rank 16	0.132	0.155
Rank 8, eight centroids	0.135	0.165

4.8 Production index

The reference serving index caches projected tokens as FP32, FP16, BF16, or symmetric per-rank INT8. It scores all chunks for a query batch in one tensor operation, applies stable top- k , accounts for masks and quantization scales, and appends new chunks without rebuilding old rows. Eight-centroid modes perform deterministic clustering offline. Backing native K/V is unchanged and is materialized only after routing.

On the GTX 950M, rank-16 FP16 occupies 1,056 bytes per 32-token chunk including its mask, versus 2,080 bytes for FP32. INT8 uses about 545 bytes; INT8 with eight centroids uses about 137 bytes. Warm single-query search is 1.5–2.1 ms and an eight-query batch sustains roughly 2,660–4,090 queries/s. Kernel-launch overhead dominates at this cohort size, so quantization primarily reduces resident index memory rather than latency. FP16 and INT8 preserve the seed-11 FP32 selections exactly in aggregate (0.118/0.230 Hotpot/QASPER recall); centroid modes trade recall for residency. Deterministic centroid construction remains expensive offline: roughly 0.7 s per Hotpot example and 2.0 s per QASPER example. Append updates average below 0.21 ms in every mode except one warm-up-tainted Hotpot FP32 mean. These are component measurements, not end-to-end serving throughput.

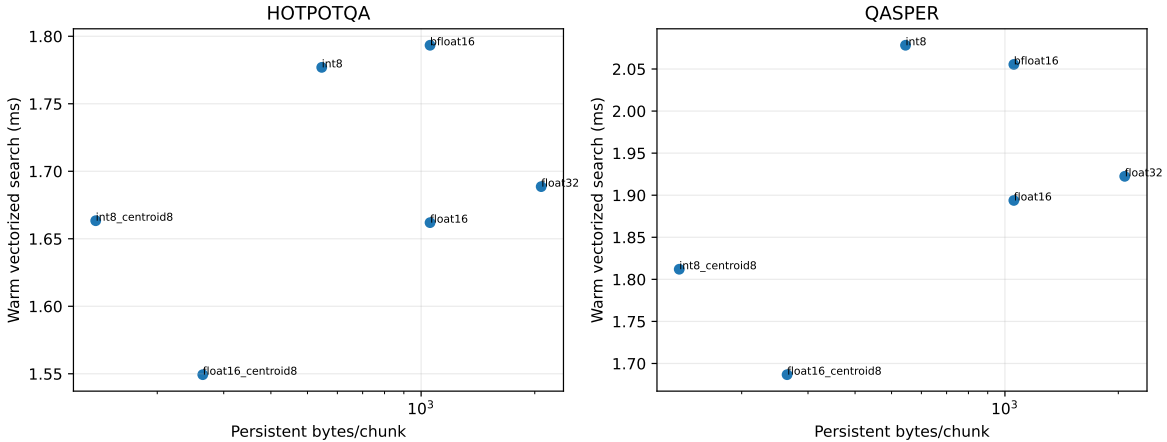


Figure 9: Warm vectorized search versus persistent routing-index bytes. Quantization changes residency far more than latency on this small, old GPU.

4.9 Comparison with Paper 2.6

Figure 10 compares the same frozen identities and four-chunk budget. Learned QK $m = 8$ is not a new overall frontier. On HotpotQA its 0.124 recall is below Paper 2.6 BM25 (0.398), exact (0.287), and iterative hybrid (0.279), with paired intervals excluding zero against all three. On QASPER it reaches 0.139: above Paper 2.6’s semantic gist (0.038) by 0.1010, with paired interval $[0.00004, 0.2270]$; above iterative hybrid (0.081) by 0.0579, interval $[-0.0610, 0.1974]$; and below exact routing (0.178) by 0.0387, interval $[-0.1951, 0.1334]$. The positive, defensible claim is dataset-specific: native response distillation improves the old QASPER semantic-gist baseline, but does not displace lexical exact matching and does not transfer to Hotpot.

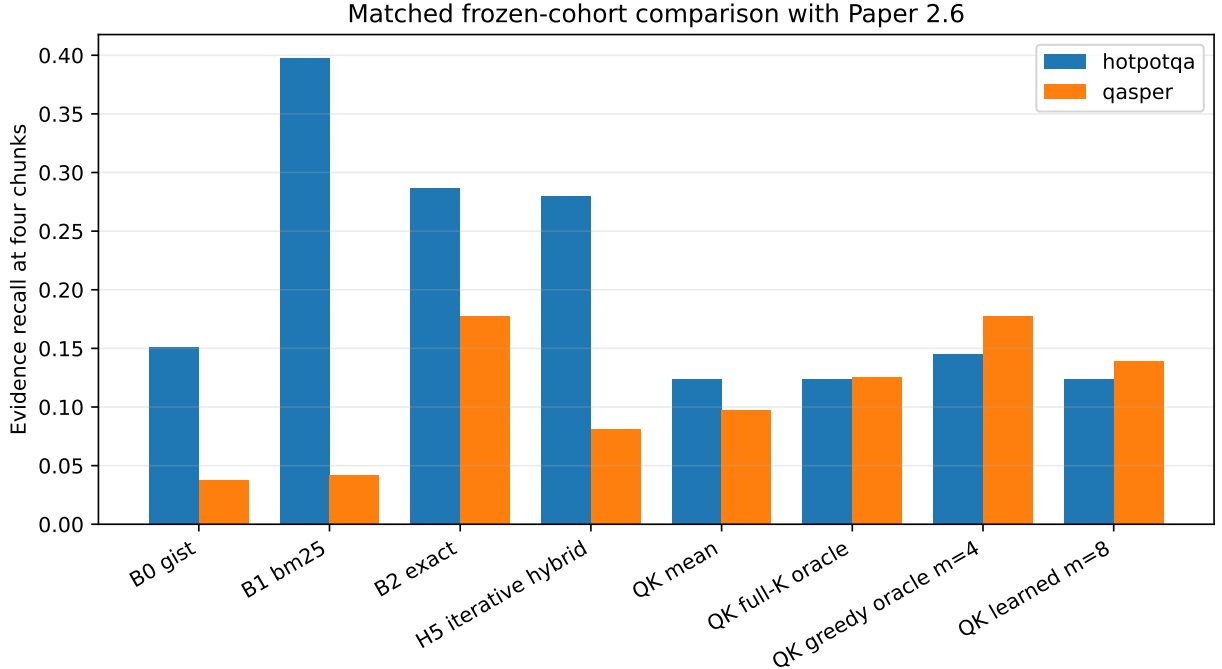


Figure 10: Matched Paper 2.6 comparison. Full-K and greedy rows are non-deployable upper bounds. Learned QK is competitive on QASPER but not Hotpot.

Paper 2.5’s native-QK closure used 256-token parents and a 20% final-parent budget. At that setting, its top-four native closure reached chain completion 0.125 on HotpotQA and 0.875 on QASPER, compared with one-shot parent values 0.138 and 0.938. Those rows concern associative successor discovery, not one-shot 32-token ranking. We preserve them as historical controls but do not pool their metrics with Table 1.

4.10 Cost

The synchronized CUDA audit includes uncached controller construction, compact-key gathering, and QK scoring. Table 8 reports means over all 32 test identities on an NVIDIA GTX 950M. Absolute times are hardware-specific; the ordering and native-dot counts expose algorithmic differences.

The learned selector is 4.5–5.0 times faster than farthest-first and about 237–271 times faster than the top- r greedy oracle, but 16.9–24.4 times slower than mean on this unoptimized Python/CUDA path. Greedy cost is dominated by query-aware candidate search, not compressed scoring. The quality CSV’s `compression_ms` field times compact assembly after indices are cached; Table 8 instead uses the separate synchronized uncached audit.

Table 8: End-to-end routing-controller cost at $m = 8$. Mean uses $m = 1$. The full-K row performs no selection.

Method	HotpotQA (50.1 chunks)		QASPER (142.6 chunks)	
	ms	native dots	ms	native dots
Mean	6.1	802	10.7	2,282
Full-K	25.9	24,898	34.1	70,853
Farthest-first	463.1	6,416	1,308.4	18,256
Greedy oracle	24,367.6	6,416	70,376.0	18,256
Learned selector	102.8	6,416	259.9	18,256

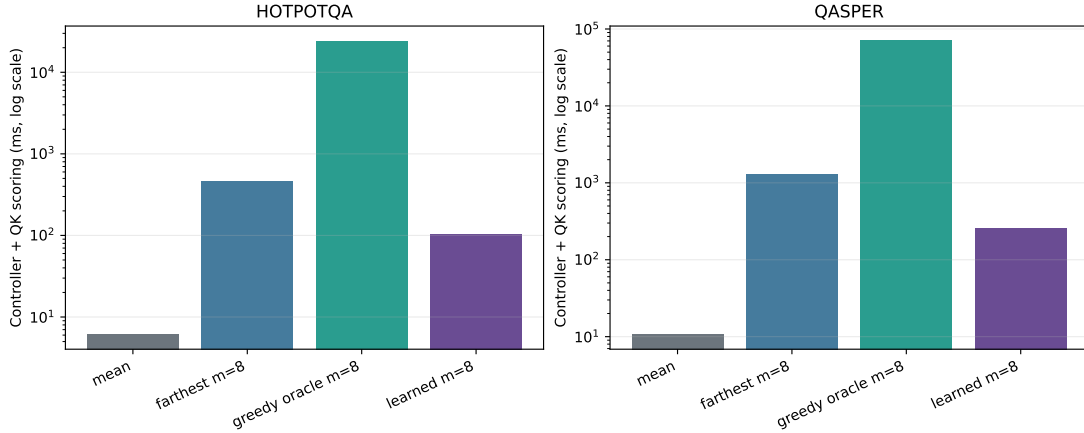


Figure 11: Controller construction plus QK scoring on a log scale. The oracle is a feasibility instrument, not a serving method.

The 2Wiki/MuSiQue extension adds a separate 20-example-per-dataset audit of one frozen seed-11 production index. It separates cold index construction from ten synchronized warm lookups; source encoding and native-K/V materialization are excluded. Rank 16 averages 1.44/1.65 ms warm routing on 2Wiki/MuSiQue, below native mean at 1.92/2.62 ms and exhaustive Python lexical scoring at about 12.8/21.0 ms. These component times do not establish end-to-end throughput. The token-native sidecar remains an exhaustive Python reference implementation, whereas low-rank search is one vectorized tensor operation.

The persistent FP32 rank-16 index, including masks, occupies 2,080 bytes/chunk; rank-8/eight-centroid occupies 264 bytes; native mean occupies 4,097 bytes. Rank-16 construction is 2.46/3.95 ms per example. Deterministic centroid construction is offline but expensive at 0.71/1.22 seconds. The four-chunk rows transfer approximately 0.50 MB of unchanged native K/V, while backing K/V averages 7.15/10.35 MB in the fixed cost cohort. Routing-index compression therefore changes address residency and search cost, not the backing-memory obligation.

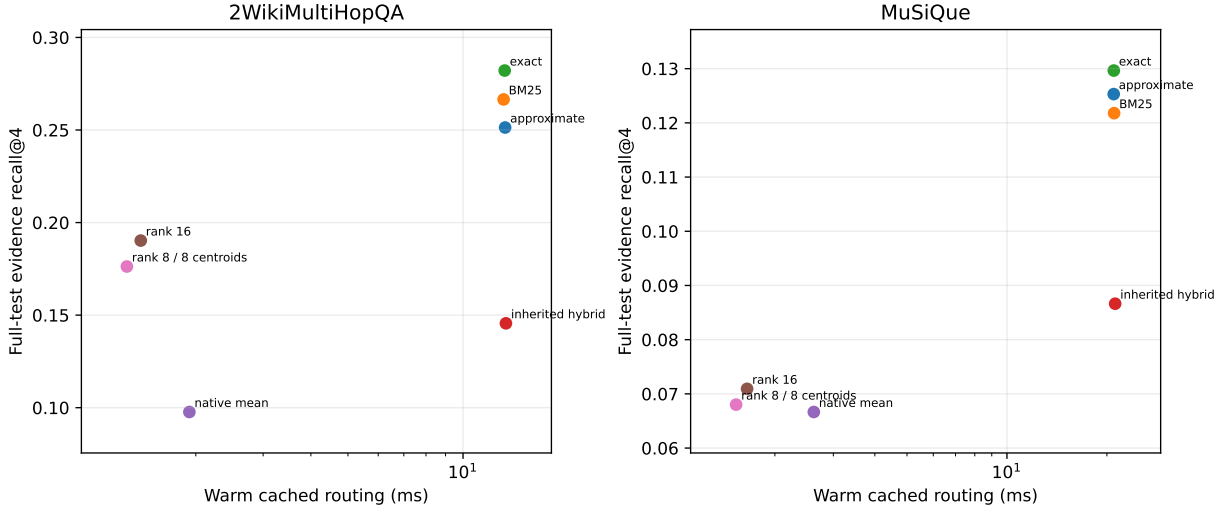


Figure 12: Full-test quality against synchronized warm component latency. Latency uses 20 fixed identities per dataset and one seed-11 learned index; quality uses all 100 held-out identities and the five-seed score ensemble. The axes therefore compare measured components, not an ensemble serving configuration.

4.11 Gate summary

Table 9: Prespecified decisions. The sequence stops at G3.

Gate	Status	Evidence
G0	Pass	224 inherited Paper 2.6 rows reproduce with zero aggregate mismatches.
G1	Pass	Controlled recall 1.00 vs. 0.95; full-K improves QASPER mean-key recall by 0.0276.
G2	Pass	Greedy $m = 8$ top-four overlap +0.3438; greedy $m = 4$ QASPER recall +0.0802.
G3	Fail	Key-only recovers 40.5%; the best query-conditioned oracle-imitation row recovers 47.3%. Both are below 80%.
G4	Closed	Synthetic off-manifold slots require G3.
G5	Closed	Streaming recurrent memory requires an offline compressor pass.

Table 10: Post-G3 diagnostic decisions. These extension gates do not reopen or revise G3.

Gate	Status	Evidence
E0	Pass	Frozen rows, hashes, model revision, feature caches, and identity partitions reproduce.
E1	Pass	Query/response-aware selectors are not explained by matched native geometry alone.
E2	Inconclusive	The prespecified query-conditioned controller improves Hotpot but regresses QASPER; a test-selected row cannot pass a confirmatory gate.
E3	Exploratory pass	Decision-aware training exceeds oracle imitation by 0.0300/0.0488 recall on Hotpot/QASPER.
E4	Pass	Direct low-rank routing retains the measured retrieval gain with at least $32\times$ less state than the full token-key stream.
E5	Pass	Rank-8, $m = 8$ static compression exceeds the old QASPER semantic gist at 256 bytes per chunk.
E6	Reopened separately	A later frozen 64+64 confirmation validates the QASPER-over-mean effect and permits the reported SmolLM portability test; it does not retroactively pass G3.

Table 11: Cross-dataset extension decisions. These gates concern the frozen low-rank architecture family and do not revise G3.

Gate	Status	Evidence
M0	Pass	240 identities, exact evidence spans, cache hashes, bounded source calls, and disjoint partitions audit cleanly.
M1	Pass on 2Wiki	Zero-shot/retrained rank 16 improve over mean by .033/.093 with intervals excluding zero; MuSiQue is unresolved.
M2	Pass on 2Wiki	Rank-8/eight-centroid retains 84.9% of rank-16 gain at 256 FP32 bytes/chunk excluding masks.
M3	Fail	Validation-selected lexical-low-rank fusion regresses on both held-out cohorts.
M4	Pass	The unchanged rank-16 architecture family is useful on QASPER and 2Wiki, with dataset-specific weights.
M5	Eligible on 2Wiki	M1 opens a future native-K/V generation replay; this routing extension does not execute it.

5 What the result means

Five observations explain the outcome.

First, a full-key teacher is not automatically an evidence oracle. Native attention keys encode many functions besides document retrieval, and a question’s final layer-27 query can respond to syntactic, discourse, and answer-format features. Top-four mean is the best validated reduction here, but its natural advantage over a mean key is small.

Second, response approximation and top-four selection are different objectives. Greedy $m = 8$ nearly reproduces the teacher ranking on HotpotQA yet does not improve recall. On QASPER, $m = 4$ retrieves more evidence than $m = 8$ despite lower teacher overlap. Additional landmarks can faithfully

restore teacher preferences that are irrelevant or harmful to evidence selection. A future objective should combine teacher distillation with validation-only ranking calibration rather than treating the teacher as ground truth.

Third, query conditioning addresses the key-only selector’s missing information need, but does not remove objective mismatch. The combined loss gives the best HotpotQA result and a poor QASPER result; decision-aware training gives the best QASPER result; oracle imitation gives the best response preservation. Capacity alone is not the explanation: increasing rank and landmark count is not monotone, and a controller can reproduce the teacher while retrieving less evidence.

Fourth, direct projection changes the problem. Instead of choosing a few full-width keys and asking them to imitate a native response, it learns a compact interaction space in which evidence chunks rank highly. The QASPER gain survives a 64-fold reduction from the full native-key stream at rank 16, and an eight-centroid rank-8 index preserves a smaller gain at 256 bytes per chunk. Yet low teacher overlap and the Hotpot failure mean this is task-adapted routing, not general native-response distillation. The result supports low-rank native features as one channel inside a selector, not as a replacement for exact, lexical, or hybrid discovery.

Fifth, architecture transfer and channel fusion are different claims. The same low-rank interaction family gives resolved gains on QASPER and 2Wiki and even a smaller zero-shot 2Wiki gain, but it remains below exact routing and does not help MuSiQue. Its selected chunks contain unique evidence, yet a static convex fusion overfits 20 validation identities and spends the fixed budget less effectively on test. A reusable semantic channel therefore still needs a calibrated per-example channel selector; complementarity alone does not provide one.

The confirmations leave two next questions rather than one. For QASPER, routing transfers across cohorts and model families while generation controls identify frozen memory integration as the limiting stage; bounded multi-layer consumption should be trained with shuffled and bottom-ranked controls retained. For cross-dataset routing, larger calibration cohorts must test channel selection before another fusion architecture. Synthetic slots and recurrent state remain premature because neither addresses these demonstrated bottlenecks.

6 Relation to prior PRA results

Paper 1 established mean-gist routing and selected native K/V. Paper 2.5 tested iterative associative closure and showed that local/native propagation can solve controlled bridges without yielding a uniform natural gain. Paper 2.6 demonstrated strong channel asymmetry: lexical evidence is particularly effective for Hotpot and exact matching for QASPER. Paper 2.7 moved structure discovery to the query side and found that better natural facet recovery could still reduce retrieval.

Paper 2.8 does not rename those mechanisms. It changes the memory-summary objective. Its HotpotQA result does not overturn Paper 2.6: lexical relational discovery still dominates a native response summary. The post-G3 rank-16 router improves further on QASPER while using half the routing state of one full-width mean key, but it is evidence-supervised and exact remains a strong control. The fresh extension narrows the relation further: low-rank native interaction transfers to 2Wiki, including zero-shot, while exact remains stronger; it does not transfer to MuSiQue. Paper 2.6’s channel asymmetry is therefore replicated rather than erased. Query-graph facets remain a query-side control and are not used to train the memory selector.

7 Limitations

The fresh Hotpot/QASPER confirmation has 128 identities, and the transfer extension adds 200 held-out 2Wiki/MuSiQue identities. Even so, the extension retains one small frozen Qwen model, one late routing

layer, one 32-token chunk size, and one primary four-chunk budget. Five seeds quantify projection initialization, while identity-bootstrap intervals quantify cohort variation; neither establishes broader domain generalization. The 20-example validation split per new dataset is small enough for task-specific projections and channel weights to overfit. Secondary six/eight-chunk rows are fixed from validation evidence-span feasibility and must not be read as a new universal budget. The controlled suite contains only 20 schematic cases.

The teacher does not include post-RoPE distance, value utility, or layer combinations. Greedy oracle landmarks see the evaluation query and cannot be deployed; evidence-supervised projections use validation labels and are task-adapted rather than unsupervised compression. Native-K/V generation uses a single frozen late layer and 12 greedy tokens; its negative QASPER result does not test trained memory integration, multiple consumption layers, or larger instruction models. Direct low-rank indexes reduce routing state, not backing native K/V. Runtime uses eager PyTorch on an old mobile GPU; INT8 is dequantized for scoring rather than using a fused integer kernel, and centroid construction is an offline Python reference path.

8 Conclusion

Compact native-key response imitation fails as a universal objective, but evidence-supervised low-rank native interaction yields a reproducible QASPER routing channel and a second, weaker 2Wiki regime. On 64 fresh QASPER identities, rank 16 improves recall over a mean key by 0.141 and transfers to a Llama-family model. On 100 fresh 2Wiki identities, zero-shot and retrained gains are 0.033 and 0.093; a 256-byte rank-8/eight-centroid index retains 84.9% of the latter gain. The mechanism does not improve MuSiQue, exact/BM25 remain stronger, and static fusion overfits. Native-K/V controls sharpen a separate boundary: informed memory can beat shuffled or bottom-ranked memory while a frozen late-layer consumer still underperforms no memory. Routing, channel selection, and reading must therefore be trained and evaluated separately. The justified next steps are larger channel-calibration cohorts and bounded multi-layer memory integration, not a claim that low-rank QK routing replaces lexical discovery.

A Reimplementation map

The implementation separates mechanism from experiment policy.

- `src/prahf/qk_compression.py` defines GQA head mapping, full-key teachers, landmark gathering, no-training selectors, the key-only and query-conditioned selectors, four training losses, response metrics, and routing metrics.
- The Paper 2.5 feature precomputer, `precompute_native_qk_features.py`, now supports identity-disjoint validation/test offsets while retaining the historical default.
- `experiments/paper2.8_qk_compression/run_gated_study.py` selects the teacher on validation, runs gates in order, trains five seeds only after G2, and emits row-level artifacts.
- `run_cost_benchmark.py` times uncached controller construction separately from cached quality evaluation; `summarize_results.py` builds matched Paper 2.5/2.6 tables and plots.
- `run_query_conditioned_study.py` executes the 120-controller matrix with durable row-level checkpoints; `summarize_query_conditioned.py` computes paired comparisons, response recovery, stability, and plots.

- `run_low_rank_frontier.py` evaluates native geometry and trains direct native- Q/K projections; `run_selector_ablation.py` isolates salience and bilinear terms; `summarize_low_rank_frontier.py` emits paired effects, seed stability, normalized cost frontiers, plots, and E0–E6 decisions.
- `src/prahf/multihop_routing_data.py` reconstructs the frozen 2Wiki/MuSiQue identities with unique evidence markers. `precompute_multidataset_native_qk.py` captures resumable frozen features; `run_multidataset_extension.py` runs the three transfer regimes and M0–M5; `run_multidataset_cost_audit.py` separates cold construction from warm routing on a fixed representative checkpoint.
- Confirmation selection uses `run_confirmation.py`. It freezes and replays five-seed selections on the offset-24 cohort.
- Generation replay uses `run_confirmation_generation.py`. It maps exact chunk identities to unchanged native K/V and emits causal controls with crash-safe row checkpoints.
- `run_cross_model_replication.py` retrains the same projection contracts on SmolLM2/Llama native geometry; `run_production_index_benchmark.py` measures quantized vectorized serving, index residency, and append cost.
- `LowRankRoutingIndex` in `src/prahf/qk_compression.py` owns only projected routing tokens, masks, and optional INT8 scales. It never owns or rewrites backing native K/V .

The core serving path is:

```
q = native_q_projection(frozen_query_state)      # [Hq, Dh]
K = cached_pre_rope_keys                        # [C, T, Hkv, Dh]
x = key_only_features(K)                       # [C, T, 8]
z = selector(x).masked_fill(~token_mask, -inf)  # [C, T]
landmarks = gather(K, topk(z, m=8))             # [C, 8, Hkv, Dh]
scores = top4_mean(q @ landmarks / sqrt(Dh))    # [C]
chunks = stable_topk(scores, k=4)
materialize_native_kv(chunks)                   # unchanged 32-token K/V
```

The query-conditioned path replaces the fourth line with `z = selector(x, q)`. The oracle substitutes a query-aware greedy search and must not be used online.

The direct low-rank serving path avoids full native-width online dots:

```
Zk = cache(Wk @ flatten_native_keys(K))        # [C, T, r]
zq = Wq @ flatten_native_query(q)              # [r]
scores = chunk_reduce(Zk @ zq / sqrt(r))        # [C]
chunks = stable_topk(scores, k=4)
materialize_native_kv(chunks)                   # unchanged K/V
```

Static joint compression replaces each chunk’s T projected tokens with four or eight deterministic centroids, medoids, or farthest representatives. Only the routing index changes; the selected chunk still contributes all of its original native K/V .

B Artifacts

Tracked results live under `docs/papers/shared/results/paper2.8_qk_compression/`. The directory contains natural and synthetic row CSVs, teacher validation, paired effects, changed-selection audits,

matched Paper 2.6 and historical Paper 2.5 controls, selector checkpoints, synchronized cost rows, plots, gate decisions, and manifests. The `query_conditioned/` subtree adds all 120 controller runs. The `low_rank_frontier/` and `selector_ablation/` subtrees add 3,776 frontier rows, 1,280 ablation rows, 15 projection checkpoints, paired identity bootstraps, seed statistics, byte/latency accounting, plots, and extension decisions. The `multi_dataset/` subtree adds evidence-alignment audits, five-seed zero-shot/retrained/mixed rows, changed-selection and paired-effect tables, compact checkpoints, M0–M5 decisions, and a separate synchronized cost audit. Large frozen feature caches remain ignored; manifests retain their hashes and exact regeneration commands. The original full test cache hash is `edc589cc...75993ae8`; validation is `87b85b25...7533cf8`.

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